Name:			Wafer Number:
Sequence	**Step**	**Equipment**	**Parameters**
0	START		100 mm silicon <100>
1	SCRIBE	scribe	each student will have a wafer number
2	MEASURE	4PP	sheet resistance and uniformity use the WNF polar recipe with 25 points
3	OXIDIZE	PECVD	~100 nm to ~500nm PECVD2 recipe WNF SiO2 350 P1 deposition rate is ~1nm/second
4	MEASURE	Filmetrics	Oxide thickness and uniformity
5	HMDS	HMDS Oven	Standard
6	SPIN	SPIN5	Material: AZ1512, static dispense step0: Dsp_Type 4, Dsp_Time 0.1 step1: ramp 1.0, dwell, 2, RPM 500 step2: ramp 0.5, dwell 45, RPM 2000
7	BAKE	SPIN5	100 C, 1 min
8	EXPOSE	AB-M	3.5 s in ABM, use the Via mask or Via and etch mask, layer 1 or layer 4, it is the one with the EE symbol at the top center
9	DEVELOP	BATH5-Develop	1 min AD10, rinse, dry
10	INSPECT	Microscope	Check resolution marks
11	MEASURE	Profilometer	resist thickness Use the dektak, the box for measuring the thickness should be right under the stylus where it comes down, the other box is across the wafer,
12	ETCH	ICP-F or Vision	ICP: MFF SiO2 Etch C4F8/O2 (~450nm/min) Vision: WNF SiO2 Etch CF4/CHF3 RF (~35nm/min)
13	MEASURE	Profilometer	resist thickness and etch depth dektak, same boxes as before
14	STRIP	BATH6-Strip/Barrel	10 min EKC, cascade, SRD, 5 min at 125 W in barrel etcher or YES asher
15	MEASURE	Profilometer	oxide thickness and etch depth calculate selectivity
16	SPIN	SPIN5	Material: NR9-3000PY, static dispense step0: Dsp_Type 4, Dsp_Time 0.1 step1: ramp 1.0, dwell, 5, RPM 500 step2: ramp 0.5, dwell 45, RPM 2000
17	BAKE	SPIN5	150 C, 1 min
18	EXPOSE	AB-M	mask: metal or layer 2 (it is the brightfield mask) 30 s, align using the keyholes and then the alignment marks labeled Via Metal and 1-2
19	PEB	SPIN5	100 C, 1 min
20	DEVELOP	BATH5-Develop	40 s, 4:1 AD10:H2O, no agitation, rinse, dry
21	INSPECT	Microscope	check alignment and resolution
22	DESCUM	Barrel Etcher	50W 30 seconds descum
23	EVAP	EVAP2	Ion gun, Al/Cr (500 nm/50 nm)
24	SPUTTER	SPUT1	Al on back of wafer
25	LIFT-OFF	BATH6-Strip	Acetone soak 24 hours, agitation or sonication, scrubbing if necessary

26	MEASURE	Profilometer	check metal thickness dektak: the metal box will be next to the etch box from before
27	INSPECT	Microscope	check liftoff resolution

Sequence	Step	Equipment	Parameters
28	MEASURE	Profilometer	
29	INSPECT	Microscope	
30	TEST	ProbeStation	capacitance, resistance, diode curves